

Use Case Name:	Manage Library Catalog	
Scenario:	Librarian maintains the catalog database, while Students and Faculty search and view library resources.	
Triggering Event:	User accesses the LIRA portal to find academic materials, or a Librarian needs to update the inventory.	
Brief Description:	This use case allows users to search, filter, and view details of library materials. It also enables the Librarian to add, edit, or delete records to maintain the catalog database. It interfaces with the Koha ILS for data retrieval.	
Actors:	Librarian, Student/Faculty, Koha ILS (External System), Microsoft Entra ID (External System)	
Related Use Cases:	Includes: <i>Manage Borrowed Books</i>	
Stakeholders:	• Student/Faculty: To easily find and view required resources. • Librarian: To maintain accurate library records. • System Administrators: To ensure database integrity.	
Preconditions:	The user is connected to the LIRA interface, and the catalog database is online. The Librarian must be authenticated and authorized for administrative access.	
Postconditions:	The requested catalog items are displayed to the user, or the catalog database is successfully updated, saved, and synchronized with Koha ILS.	
Flow of Activities:	Actor	System
	1. Student/Faculty or Librarian accesses the portal and initiates sign-in.	1.1 Redirects authentication request to Microsoft Entra ID.
	2. Microsoft Entra ID authenticates the user credentials.	2.1 Verifies authentication and grants access to the appropriate interface based on user role.
	3. Student/Faculty enters search criteria (e.g., title, author) into the catalog interface.	3.1 Queries local database records and the Koha ILS to retrieve matching items. 3.2 Displays a list of matching catalog items.

Use Case Name:	Manage Library Catalog <i>(Continued)</i>	
Flow of Activities <i>(Continued):</i>	Actor	System
	4. Student/Faculty selects a specific item to view detailed information and availability.	4.1 Displays detailed information for the selected item.
	5. Librarian accesses the catalog management dashboard.	5.1 Displays the current catalog list and data entry forms.
	6. Librarian submits new item details or confirms an item deletion.	6.1 Validates the submitted data, updates the catalog database, and synchronizes with the Koha ILS. 6.2 Displays a success confirmation message.
Exception Conditions:	• 3.1 If no items match the user's search criteria, the system displays a "No results found" message. • 6.1 If data validation fails during an update, the system highlights errors and prompts the Librarian to correct them.	

Use Case Name:	Manage Borrowed Books	
Scenario:	Students and Faculty check out available books, while the system records and tracks the circulation status.	
Triggering Event:	User decides to borrow a specific catalog item while viewing its details.	
Brief Description:	Handles the checkout process for a user. It records the transaction by communicating with the Koha system to manage the borrowed status and due dates.	
Actors:	Student/Faculty, Koha ILS (External System)	
Related Use Cases:	Extended from: <i>Manage Library Catalog</i>	
Stakeholders:	• Student/Faculty: To successfully acquire a book. • Koha ILS: To track inventory lifecycle and circulation.	
Preconditions:	The user possesses an active, valid library account. The selected item is available in inventory. The LIRA system is connected to the Koha API.	
Postconditions:	The item is marked as borrowed, the user's account reflects the loan, the transaction is logged in Koha, and a due date is assigned.	
Flow of Activities:	Actor	System
	1. Student/Faculty submits a borrow request for a selected catalog item.	1.1 Verifies the user's account status and current borrowing limits. 1.2 Prompts the user for final confirmation.
	2. Student/Faculty confirms the borrow request and accepts the terms.	2.1 Transmits the transaction data to Koha ILS to log the checkout.
	3. Koha ILS confirms the transaction log is committed.	3.1 Updates the local item status to 'Checked Out'. 3.2 Displays a borrowing confirmation receipt with the due date to the user.
Exception Conditions:	• 1.1 If the user has a blocked account or overdue items, the system denies the borrow request and displays a warning. • 2.1 If the item becomes unavailable during the transaction, the system cancels the process and notifies the user. • 2.2 If Koha ILS is unreachable, the system queues the transaction locally and alerts the Librarian.	

Use Case Name:	Manage Chatbot	
Scenario:	Students and Faculty ask the AI questions for research assistance, while the Librarian configures the AI's knowledge base.	
Triggering Event:	User submits a query to the chat widget, or a Librarian accesses the AI configuration panel.	
Brief Description:	This use case encompasses both the automated generation of responses by the AI actor for user queries, and the administrative tuning of the chatbot's parameters and knowledge sources by the Librarian.	
Actors:	Student/Faculty, Librarian, AI Engine	
Related Use Cases:	N/A	
Stakeholders:	• Student/Faculty: For immediate research assistance. • Librarian: To ensure the AI provides accurate answers and reduce manual reference queries.	
Preconditions:	The AI engine is online and accessible. The Librarian must be authenticated to access the configuration panel.	
Postconditions:	A relevant response is generated for the user, or the chatbot's configuration and knowledge base is successfully updated.	
Flow of Activities:	Actor	System
	1. Student/Faculty submits a question into the chat widget.	1.1 Parses the query and routes it to the AI Engine.
	2. AI Engine generates a synthesized response based on the prompt and returns it.	2.1 Formats the retrieved answer, displays it in the chat interface, and logs the interaction.
	3. Librarian accesses the chatbot configuration panel.	3.1 Displays current knowledge sources and prompt settings.
	4. Librarian updates custom prompts or uploads new knowledge documents.	4.1 Validates and applies the new configurations to the AI engine.
Exception Conditions:	• 1.1 If the AI service times out, the system apologizes and provides contact information for a human librarian. • 2.1 If the AI generates a response violating safety policies, the system intercepts it and substitutes a safe default message. • 4.1 If the Librarian uploads an incompatible document, the system rejects it and displays an error message.	

Use Case Name:	Manage Reports	
Scenario:	The Librarian generates analytics regarding library usage, circulation data, and chatbot interactions.	
Triggering Event:	Librarian requests a statistical or operational report from the dashboard.	
Brief Description:	Allows the Librarian to compile and export data reports by extracting data from local LIRA databases and integrating circulation data from the Koha ILS.	
Actors:	Librarian, Koha ILS (External System)	
Related Use Cases:	N/A	
Stakeholders:	• Library Management: To make data-driven decisions regarding acquisitions and system usage.	
Preconditions:	The Librarian holds administrative privileges. Log data exists in both the LIRA and Koha databases.	
Postconditions:	A formatted report is generated and is available for viewing or download.	
Flow of Activities:	Actor	System
	1. Librarian selects a report type (e.g., circulation stats, AI usage) and date range from the dashboard.	1.1 Requests necessary usage data from the local database and circulation data from Koha ILS.
	2. Koha ILS provides the requested raw circulation data.	2.1 Aggregates the combined data and displays a report preview on the dashboard.
	3. Librarian clicks to export the final report.	3.1 Generates a downloadable file (PDF/Excel) and provides it to the user.
Exception Conditions:	• 1.1 If Koha ILS is offline, the system generates a partial report using only local LIRA data and appends a warning note. • 2.1 If the selected date range contains no data, the system outputs an empty report accompanied by a notification.	